

CS3 Rubric – Skin Lesion Image Analysis

Thomas Hooks

DS 4002 – Spring 2026 - Instructors: Javier Rasero; TA: Ethan Meidinger

Due: May 2026

Submission format:

- Upload link to GitHub repo to canvas AND
- Provide a PDF link to the finalized Presentation

Individual Assignment

General Description: Follow the instructions of this rubric and the outline in GitHub repository to produce the desired outputs. Use these outputs to finalize the GitHub repository and produce a Presentation of the process and results.

Preparatory Assignments: Review the preparatory materials in the GitHub Repository.

Why am I doing this? This project challenges you to apply machine learning to a high-stakes medical problem: skin cancer detection. You will explore how multimodal data, combining dermoscopic images with patient metadata, affects diagnostic reliability. By reframing technical results for a clinical audience, you practice the essential skill of communicating how data-driven tools can improve patient triaging and early intervention.

- Course Learning Objective: Evaluate the clinical impact of integrating metadata into image-based models.
- Course Learning Objective: Translate complex model metrics (like AUC) into actionable medical insights.
- Course Learning Objective: Prepare findings for presentation to your peers.

What am I going to do? First, you should read the Hook Document to get an overview of the clinical stakes involved in melanoma detection. Then, read the provided resources to gain a deeper understanding of how imaging aids in early detection. Next, use the README and other available resources as a guide to obtain and clean the clinical metadata and perform your image analysis. Lastly, you will prepare a presentation on how you obtained your results and any significant findings regarding the effectiveness of multimodal classification in a clinical setting. Deliverables include:

- GitHub Repository - to provide all annotated code, figures, and resources so that your diagnostic results may be replicated
- Presentation - to guide the audience through your work and clinical conclusions.

Tips for success:

- Focus on the Meaning – As a student, you need to understand why clinical metadata matters. Don't just show the code, explain how patient age or lesion site changes the diagnostic story and helps a doctor make a better decision.

- Visualize the Medical Impact – Use your output folder to store more than just graphs. Ensure your confusion matrices and AUC scores are framed in terms of patient safety.
- Prioritize Clarity over Complexity – While deep learning is complex, your explanation shouldn't be. Aim for a balanced level of detail in your explanations, technical enough to be accurate, but accessible enough for someone who isn't too experienced in machine learning.

How will I know I have Succeeded? You will meet expectations on this assignment when you follow the criteria in the rubric below.

Formatting	• Repository Link: A working link to a public GitHub repository. • File Standards: All documents (README, LICENSE, etc.) must be organized logically within the repository. • Presentation: A final PDF presentation of approximately 10 slides in length.
GitHub Repository	• Goal: This repo is your credible method of explaining your diagnostic results. Someone should be able to download this repo and produce the same findings and easily understand how to do so. • README.md: Explains repo usage with "##" sections; includes software requirements and instructions for reproduction. • LICENSE.md: Inclusion of a standard open-source license (e.g., MIT). • SCRIPTS Folder: Finalized scripts for data cleaning and model training with header comments and internal annotations. • DATA Folder: Includes the initial MILK10k metadata and the final processed version. • OUTPUT Folder: Contains tables (AUC scores) and figures (confusion matrices/training curves) for replication. • Presentation File: A copy of the finalized PDF presentation stored within the repository.
Presentation	• Goal: This presentation is how you will primarily be communicating with your peers. It will allow you to communicate your findings clearly and efficiently with a focus on your model and results. • Core Content: ○ Title/Introduction: Context of the skin lesion study. ○ Clinical Research Question: The medical "Why" behind the project. ○ Data Explanation: Insights into the MILK10k dataset and metadata. ○ Model Explanation: Details on the timm architectures and multimodal approach. ○ Analysis/Results: Comparison of performance metrics with a focus on model outcomes. ○ Clinical Implications: How this affects real-world early detection. ○ Next Steps: Future improvements or research directions.

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